**Lab-1**

1. Develop a simple rule-based expert system for animal identification. Define a set of facts and IF–THEN rules and infer the type of animal based on the given facts.

2. Develop an interactive expert system for student course recommendation. The system should ask the user questions about marks, interests, and skills and recommend a suitable course using predefined IF–THEN rules.

3. Implement a forward-chaining inference engine for an expert system. Given a set of initial facts and IF–THEN rules, repeatedly apply applicable rules to derive new facts until the required conclusion is reached or no further rules can be applied.

4. Develop an expert system for fault diagnosis using backward chaining. Given a suspected fault as the goal, identify the rules and facts required to prove or reject the hypothesis. The system should also display an explanation showing how the conclusion was reached.